MINIMIZING PEAK POST-MATCH QUEUES

Optimal Bus Dispatch Scheduling to Support Stadium Egress

at MetLife Stadium During FIFA World Cup 2026

VENUE	EVENT	MODEL	BASE CASE z*
MetLife Stadium, NJ	FIFA World Cup 2026	MILP Optimization	6,490 passengers

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AGENDA

What We'll Cover Today

A structured walkthrough of the MILP model, results, and insights for MetLife Stadium egress.

01 **Problem Background**
Operational challenge & motivation

02 **Model Formulation**
MILP structure, variables & constraints

03 **Base Case Results**
Optimal dispatch plan & queue evolution

04 **Scenario Analysis**
Low / Base / High parameter comparison

05 **Sensitivity Analysis**
Key levers driving peak queue outcomes

06 **Insights & Conclusion**
Recommendations & limitations

01 PROBLEM BACKGROUND & MOTIVATION

The Operational Challenge

- MetLife Stadium hosts 8 FIFA WC 2026 matches, including the Final (July 19)
- Soccer capacity: 82,500 spectators — largest venue in the tournament
- Primary transit route: NJ TRANSIT through Secaucus Junction
- Inbound arrivals are gradual; post-match egress is the critical bottleneck
- After final whistle, a disproportionate share of the crowd seeks transport within minutes
- NJ TRANSIT served 150,000+ combined bus/rail trips across 9 Club World Cup matches (2025)
- 20,390 outbound passengers on CWC final Sunday — a confirmed operational precedent
- Auxiliary bus service formally approved for WC 2026

Modeling Focus: Post-match bus dispatch from MetLife → Secaucus Junction. Rail = fixed; buses = optimized.

01 PROBLEM BACKGROUND & MOTIVATION

Why This Problem Matters

- **Safety & Crowd Control**

  Brief but extreme queue spikes create dangerous platform congestion — harder to manage than sustained high loads

- **Operational Precedent**

  Buses are not hypothetical — NJ TRANSIT formally approved auxiliary bus service after a successful CWC 2025 test run

- **Reputational Stakes**

  A globally visible FIFA Final leaves zero margin for transit failures under international scrutiny

- **OR Application**

  Time-varying demand + discrete fleet + finite intervals = a natural fit for MILP

Model Scope

✓ IN SCOPE

- Post-match bus dispatch from MetLife Stadium
- 12 consecutive 15-minute intervals (3-hour horizon)
- Fleet turnaround constraint modeled explicitly
- Rail capacity treated as fixed background input
- Minimax objective: minimize worst-case queue

✗ OUT OF SCOPE

- ✗ Full NJ–NY regional transit network
- ✗ Secaucus Junction downstream capacity
- ✗ Traffic dynamics / travel time variability
- ✗ Private car, rideshare, or other egress modes

Modeling Focus: Post-match bus dispatch from MetLife → Secaucus Junction. Rail = fixed; buses = optimized.

02 MODEL PARAMETERS, DECITION VARIABLES & OBJECTIVE FUNCTION

Parameters

D_t	Passenger demand in interval t
R_t	Fixed rail capacity in interval t
B	Total number of buses available
C_b	Effective capacity per bus
h	Intervals required for bus turnaround
L	Max buses loadable in one interval
Q_0	Initial queue at start of horizon

Decision Variables

x_t	Buses dispatched in interval t	*Integer ≥ 0*
y_t	Passengers served in interval t	*Continuous ≥ 0*
q_t	Queue remaining at end of interval t	*Continuous ≥ 0*
z	Maximum queue across all intervals	*Continuous ≥ 0*

Objective Function

Minimize

$$M \cdot z + \sum q_t \quad \forall t \in T$$

Term 1 — $M \cdot z$ ($M = 10^6$)

Heavily penalizes the peak queue z. The large weight M forces the solver to minimize the worst-case queue first, before anything else.

Term 2 — $\sum q_t$

Minimizes the total accumulated queue across all intervals. This secondary objective smooths the queue over time, not just at the peak.

02 FLOW OF PASSANGERES IN EACH INTERVAL

Passengers Arriving in Interval t	D_t
Total Queue to Process	$q_{t-1} + D_t$
Available Service Capacity	$R_t + C_b \cdot x_t$
Passengers Served	y_t
Remaining Queue	q_t

Core Queue Equation

New Queue

= Previous Queue (q_{t-1})

+ New Demand (D_t)

− Passengers Served (y_t)

$$q_t = q_{t-1} + D_t - y_t$$

Variable Key

D_t	Demand in interval t
R_t	Rail capacity
$C_b \cdot x_t$	Bus capacity
y_t	Passengers served
q_t	Remaining queue

02 MODEL CONTRAINTS

1 Queue Evolution

$$q_t = q_{t-1} + D_t - y_t \quad \forall t \in T$$

The queue at the end of this interval equals the previous queue plus new demand minus passengers served.

This is the most important constraint in the model — it tracks how the queue evolves over every interval.

2 Service Capacity

$$y_t \leq R_t + C_b \cdot x_t \quad \forall t \in T$$

Passengers served cannot exceed rail capacity plus bus capacity.

Rail capacity R_t is fixed. Buses x_t are the flexible, controllable part of the model.

3 Demand Availability

$$y_t \leq q_{t-1} + D_t \quad \forall t \in T$$

The system cannot serve more passengers than the number currently waiting or arriving.

This prevents the model from producing unrealistic solutions where more people are served than are actually present.

! Key Insight

Constraints 1, 2, and 3 together define the core feasibility of the model:

- The queue must evolve correctly (C1)
- Service is bounded by capacity (C2)
- Service is bounded by demand (C3)

Without all three, the model could produce solutions that are mathematically optimal but physically impossible.

Model Constraints

4 Maximum Queue Definition

$$z \geq q_t \quad \forall t \in T$$

z must be at least as large as every queue value. Because the objective minimizes z, the model will push down the largest queue — effectively minimizing peak congestion.

5 Bus Loading Limit

$$x_t \leq L \quad \forall t \in T$$

Even if many buses are available, only L buses can be loaded and dispatched in one 15-minute interval. This captures real operational loading capacity at a station.

6 Fleet Availability (Turnaround)

$$\sum x_s \leq B \quad s \in [\max(1, t-h+1), t]$$

Buses dispatched recently are still in service — they need time to complete their trip and return (turnaround time h). So active buses cannot exceed total fleet B. This makes the model realistic: without it, the same buses could be reused too quickly.

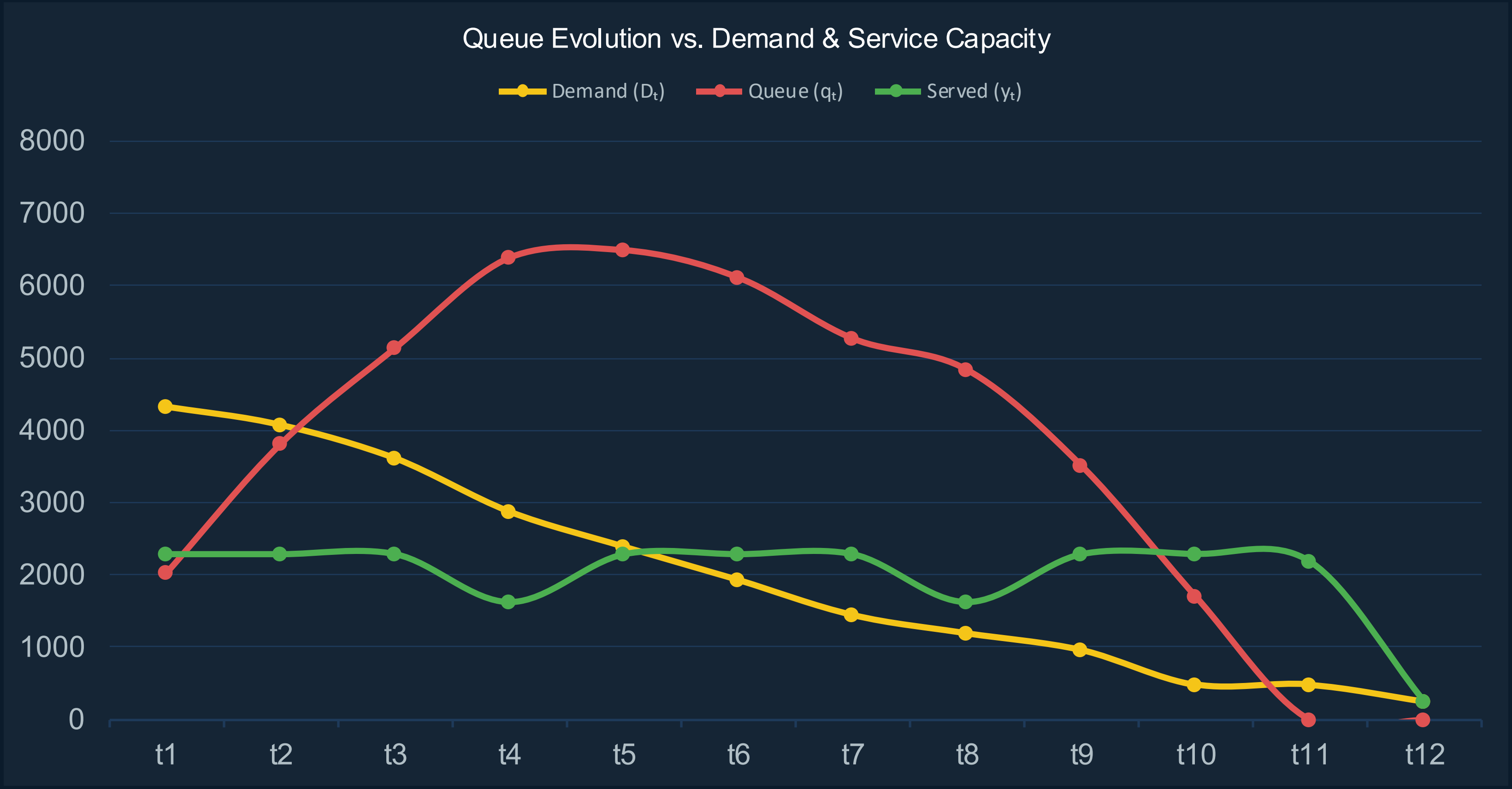
Ω Variable Domains

$$x_t \in \mathbb{Z}^+ \quad \forall t \in T \qquad y_t \geq 0, \; q_t \geq 0, \; z \geq 0$$

Key point: x_t must be an integer — we cannot dispatch 3.7 buses.
The continuous variables y_t, q_t, and z are allowed to take any non-negative value.

03 BASE CASE RESULTS

6,490	**Interval 5**	**28.3 min**	**Interval 11**	**45,290**
Peak Queue (z*)	Peak Occurs At	Avg. Wait Time	Queue Cleared	Total Queue (pass-intervals)



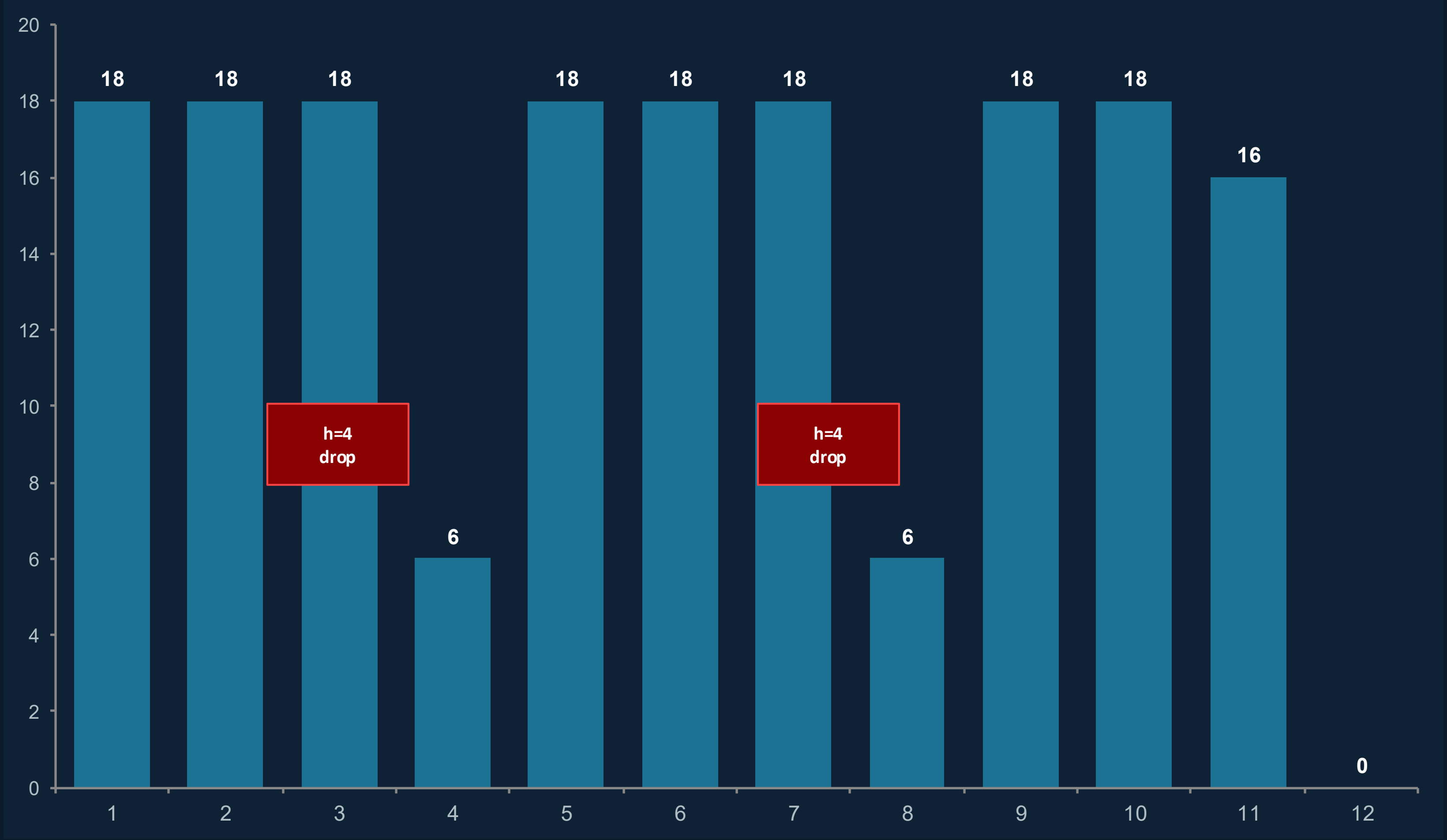
Key Findings

- Queue builds through first 5 intervals as demand outpaces combined rail + bus capacity
- Maximum allowable 18 buses dispatched in intervals 1–3
- Turnaround constraint (h=4) forces dispatch to drop from 18 → 6 buses in interval 4
- Peak queue 6,490 passengers reached at end of interval 5 (60–75 min post-match)
- Queue falls steadily after interval 5 as demand declines
- Fully cleared by interval 11; interval 12 served by rail alone

Intervals 1–12 = 15-min windows. Intervals 4 & 8 show turnaround-driven dispatch drops.

03 OPTIMAL DISPATCH PATTERN & TURNAROUND DYNAMICS

Optimal Bus Dispatch by 15-Min Interval (x_t)



Front-Loaded Dispatch

Optimizer sends maximum 18 buses in intervals 1–3. Early demand is highest; any delay raises the eventual peak.

Turnaround Constraint (h=4)

Buses dispatched in intervals 1–3 are still in transit during interval 4. Only 6 buses available → capacity drops from 2,290 to 1,630.

Same Effect in Interval 8

A second turnaround dip occurs in interval 8 for the same reason. The pattern repeats every h intervals.

Late Intervals

Interval 11: only 16 buses needed to clear remaining queue. Interval 12: rail alone handles 240 remaining passengers.

Key insight: turnaround time is a critical binding factor. Counting fleet size alone overestimates effective capacity.

04 SCENARIO ANALYSIS

Parameter	Low (Conservative)	Base Case ★	High (Enhanced)
Total Demand	20,000 pass.	**24,000 pass.**	28,000 pass.
Rail Cap / Interval	1,050 pass.	**1,300 pass.**	1,550 pass.
Fleet Size (B)	45 buses	**60 buses**	70 buses
Bus Capacity (C_β)	45 pass.	**55 pass.**	65 pass.
Turnaround (τ)	75 min (h=5)	**60 min (h=4)**	45 min (h=3)
Loading Limit (L)	12 buses/int	**18 buses/int**	24 buses/int
Peak Queue (z*)	**4,945**	**6,490**	**7,570**
Total Queue (Σq_t)	**43,875**	**45,290**	**41,810**
Total Dispatched	**114 trips**	**172 trips**	**210 trips**

Key Insight

The high scenario has a LARGER peak queue (7,570) but a LOWER total queue (41,810) than the base case.

Why? The high scenario is high-demand AND high-capability: more buses, faster turnaround, larger loading limit.

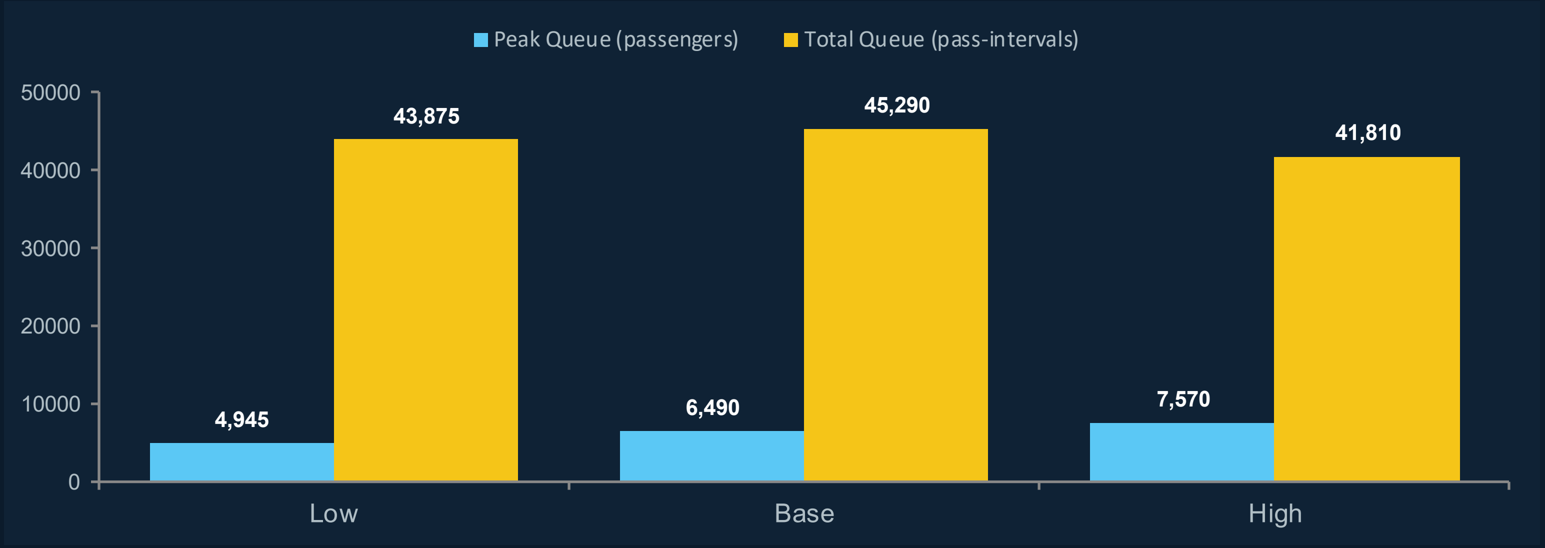
Operational enhancements can partially offset greater demand — but the worst-case spike still grows.

Takeaway: improving operations without managing demand controls the total burden, not peak congestion.

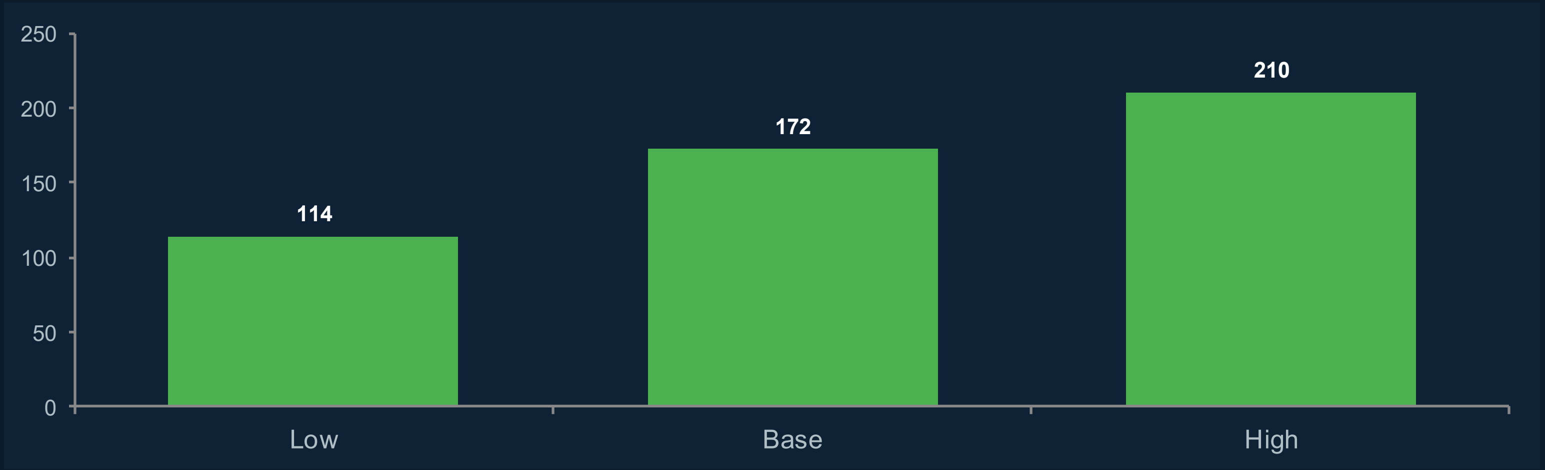
All three scenarios solve to optimality. Parameter changes interact — isolated effects require one-at-a-time sensitivity analysis (Section 7).

04 SCENARIO ANALYSIS

PEAK QUEUE vs. TOTAL QUEUE BY SCENARIO



TOTAL BUSES DISPATCHED BY SCENARIO



Key Takeaways

01

Peak Queue Rises with Demand

Higher demand = larger worst-case queue, even with enhanced operations. The High scenario shows 7,570 despite all improvements.

02

Total Burden Tells a Different Story

High scenario total queue (41,810) is lower than Base (45,290). Faster turnaround + more buses reduce overall wait time after the peak.

03

Parameters Interact — Don't Change One

Each scenario shifts 6 parameters at once. Isolating individual effects requires the one-at-a-time sensitivity analysis in Slide 7.

04

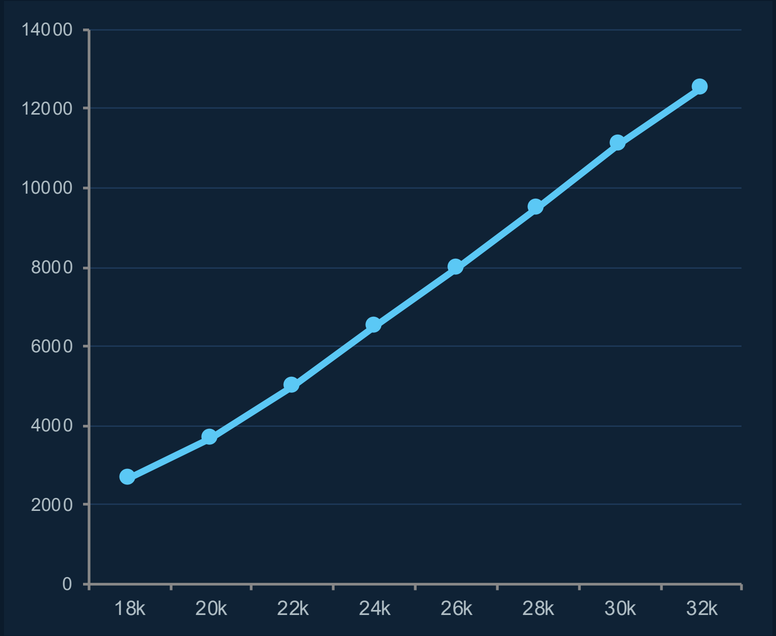
Operations Matter as Much as Demand

The gap between Low (114 buses) and High (210 buses) in dispatches shows that enhanced ops unlock significantly more throughput.

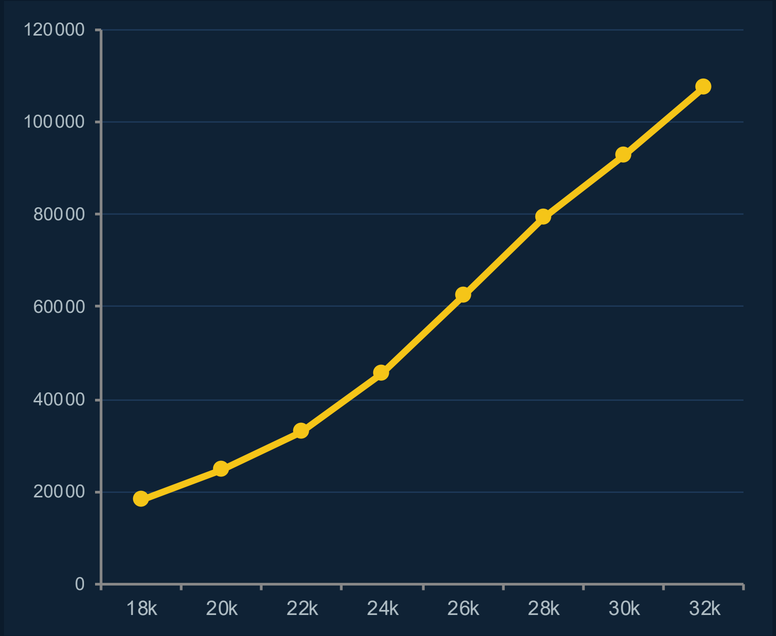
05 DEMAND SENSTIVITY ANALYSIS

NEW · All base-case operating parameters fixed — only total demand varies. Isolates demand as a standalone risk driver beyond the three scenarios.

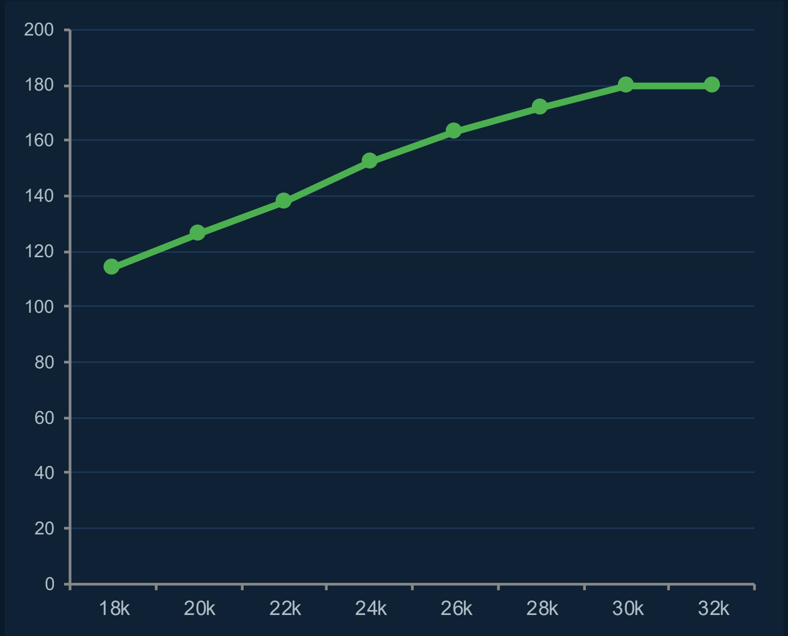
Peak Queue vs. Total Demand



Total Queue vs. Total Demand



Total Buses Dispatched vs. Demand

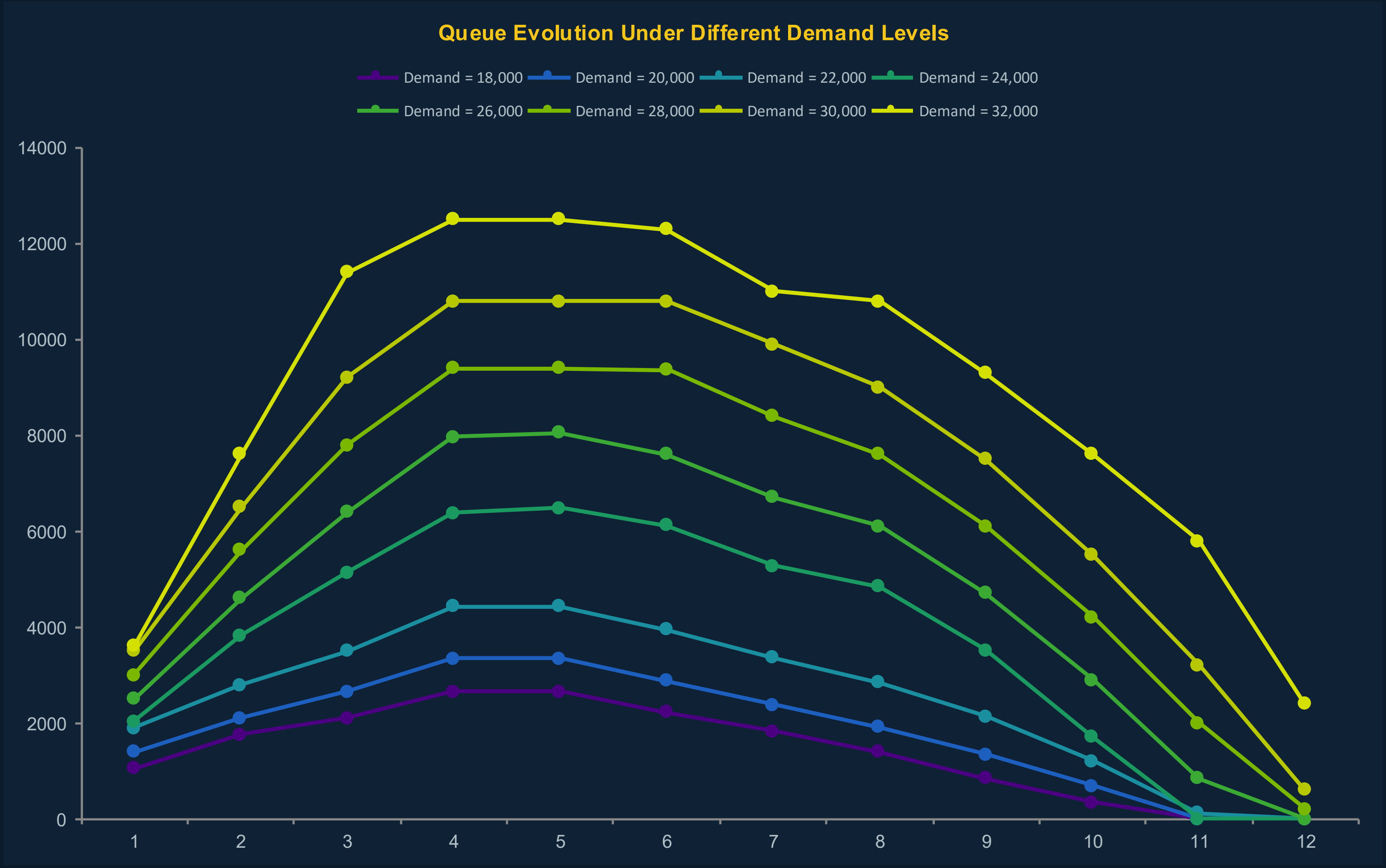


2,660	6,490	12,520	26,000
Peak queue at 18,000 pass.	Base case (24,000 pass.)	Peak queue at 32,000 pass.	Capacity ceiling ⚠

All base-case parameters fixed (60 buses, 55 pass/bus, τ=60 min, L=18, R=1300/interval). Only total demand is varied from 18,000 to 32,000.

05 QUEUE PROFILES ACROSS DEMAND LEVELS

Higher demand shifts the peak later and extends clearance time well past Interval 11



≤ 20,000 pass.

Queue peaks and clears comfortably within the 3-hour window.

24,000 pass.

Base case — peaks at 6,490, clears by Interval 11.

26,000+ pass.

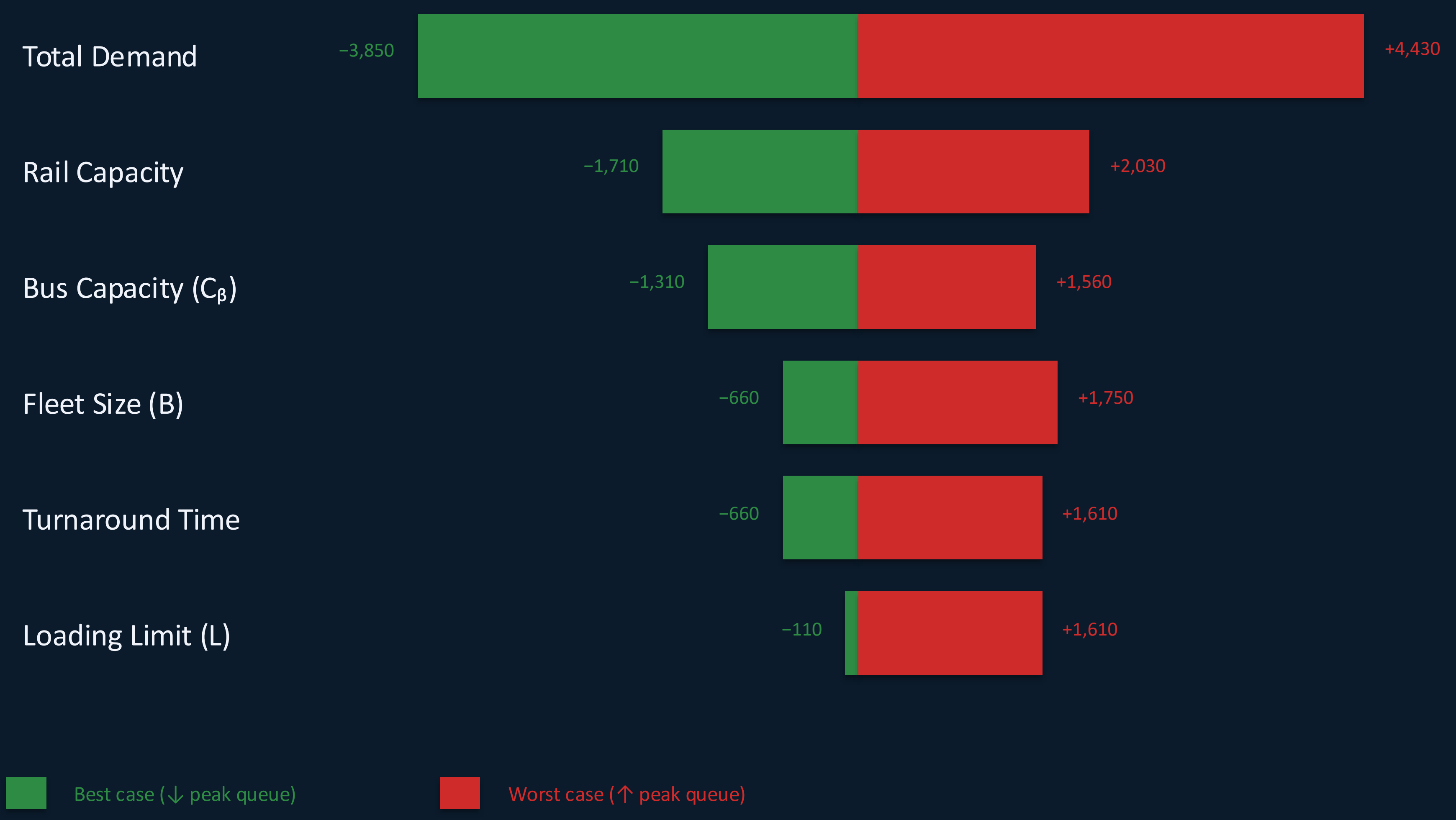
Operational ceiling: bus dispatches max at 180 total — additional demand is unabsorbed.

30,000+ pass.

Queue stays elevated through Interval 12. Demands demand management or expanded fleet.

06 SENSITIVITY ANALYSIS — ONE-AT-A-TIME (OAT)

Change in Peak Queue (z*) from Base Case (6,490)



OAT Takeaways

- Total demand is the dominant driver — a ±change of up to 4,430 passengers
- Rail capacity is foundational but cannot be adjusted on match day — plan in advance
- Bus capacity (C_β) has a multiplier effect — each extra passenger/bus amplifies every dispatch
- Fleet size & turnaround time have similar best-case impact (−660 each) but turnaround worsening is costly
- Loading limit has the smallest upside — relaxing L helps only if fleet & turnaround are not binding
- Demand management + fixed-capacity improvements are the strongest combined levers

06 TWO-WAY SENSITIVITY — FLEET SIZE × TURNAROUND TIME

Fleet Size (B)	Turnaround Time (τ) 30 min (h=2)	45 min (h=3)	**60 min (h=4)**	75 min (h=5)	90 min (h=6)
B=30	6,490	8,100	8,240	9,130	9,750
B=45	5,830	6,450	7,315	8,305	8,925
B=60	5,830	5,830	**6,490★**	7,480	8,100
B=75	5,830	5,830	5,830	6,655	7,275
B=90	5,830	5,830	5,830	5,830	6,450

Low queue (better) High queue (worse)

★ = Base case (B=60, τ=60 min, z*=6,490)

Heatmap Insights

- Faster turnaround partially compensates for a smaller fleet — 30 buses at τ=30 min matches the base case peak
- Diminishing returns from fleet expansion: B=75 and B=90 show the same peak across several τ values
- Once fleet is large enough, another constraint becomes binding (demand timing, rail capacity, or loading limit)
- Worst outcome: small fleet (B=30) + slow turnaround (τ=90) → peak queue of 9,750
- Planners should target the combination of fleet + turnaround that achieves a threshold, not maximize one lever

07 MANAGERIAL INSIGHTS & RECOMMENDATIONS

① Dispatch Aggressively Early

Pre-stage buses before the final whistle.
First 30–45 min is the critical window.
Any delay in deployment irreversibly raises the peak queue.

② Prioritize Turnaround Reduction

Use dedicated return lanes, multi-door loading, and Secaucus coordination.
Cutting τ from 60→45 min rivals adding more buses.

③ Expand Staging & Loading Capacity

Third, increasing staging or loading capacity helps but only if other constraints are also improved.
If turnaround time or fleet size is limiting, then expanding loading alone won't reduce congestion

④ Use Demand Management

Staggered egress by section, on-site entertainment, and real-time app guidance.
Our results show that shifting just 5 to 10 percent of passengers away from the first interval can noticeably reduce the peak queue

⑤ Set Acceptable Queue Thresholds

defining acceptable queue limits for safety and operations.
These thresholds can be used to trigger contingency actions when demand exceeds expectations.

⑥ More Buses Alone Is Not Sufficient

Effective capacity = f(fleet size, turnaround, loading limit).
Adding buses helps only until another constraint binds. Use OAT to identify which lever matters most.

Strongest strategy: demand control + reliable rail capacity + efficient bus turnaround + dispatch combination.

08 LIMITATIONS & FUTURE IMPROVEMENTS

! Model Scope

Secaucus Junction receiving end treated as unconstrained. Real downstream capacity could extend turnaround times.

! Traffic Dynamics

Turnaround time is fixed, not traffic-dependent. Road congestion during an 82,500-person event is likely material.

! Demand Profile Uncertainty

Departure curves are scenario-based assumptions, not empirically derived from interval-level NJ TRANSIT data.

! Single-Mode Focus

Model covers only NJ TRANSIT-route passengers. Mode-share from private cars, rideshares etc. is implicitly absorbed into total demand.

Future Extensions

- → Incorporate Secaucus Junction downstream capacity as an explicit constraint
- → Use stochastic demand profiles for risk-adjusted solutions across demand realizations
- → Extend to multi-mode joint optimization (buses + rail + park-and-ride)
- → Integrate real-time monitoring for dynamic re-optimization during the post-match period
- → Calibrate demand profiles using interval-level boarding data from comparable MetLife events

CONCLUSION

Key Takeaways

- MILP model minimizes peak post-match queue (z*) at MetLife Stadium for FIFA WC 2026
- Base case: 6,490 peak passengers, cleared by interval 11, avg wait ~28.3 minutes
- Fleet turnaround time is the most critical structural operational constraint
- Dispatch buses aggressively in the first 30–45 minutes — front-loading is optimal
- Total demand and rail capacity are the largest drivers of queue outcomes (OAT)
- Operational enhancements can reduce total waiting burden even when peak stays high
- Model serves as a practical pre-event planning & stress-testing tool

The 2026 World Cup creates unusually demanding post-match conditions. Systematic pre-event optimization using this framework supports safer, more efficient egress.

$z^* = 6,490$

Base Peak Queue

3 Scenarios

Low / Base / High

6 OAT Tests

Sensitivity Parameters

$\tau \rightarrow$ key factor

Turnaround Impact

Interval 11

Full Queue Clearance

THANK YOU

Questions?

We'd love to hear your feedback

Anh · Ashley · Bahareh · Claire · Karthik